

Module 5

React Events

Event Handling

- In React, handling events is very similar to handling event in regular JavaScript.
- Event handling allows us to respond to user actions like clicks, input and form submissions
- However, there are some syntactic differences and conventions that are important to understand.

Why Event Handling?

- Makes the app interactive.
- Responds to user actions.
- Updates state and UI.
- Brings your app to life.

How it works?

- 1 User perform an action (click or type)
- 2 Event is triggered (onClick, onChange etc)
- 3 Function runs and updates the state
- 4 UI re-renders and shows updated data

Event Binding Example

- In React, events are written in camel Case , and you pass a function as the event handler.

```
1  import React from 'react'
2
3  function MyButton() {
4    let handleClick={()=>{
5      alert("Button was clicked!!!")
6    }}
7    return (
8      <div>
9        <button onClick={handleClick}>Click me</button>
10     </div>
11   )
12 }
13
14 export default MyButton
```

Passing Arguments to Event Handlers

- Sometimes you might need to pass additional parameters to your event handler.

```
1  import React from 'react'
2
3  function MyButton() {
4    let handleClick=(name)=>{
5      |      alert(`Hello, ${name}`)
6    }
7    return (
8      <div>
9        <button onClick={()=>handleClick("React Developer")}>
10         Click me</button>
11      </div>
12    )
13  }
14
15  export default MyButton
16
```

Preventing Default Behavior

- Sometimes you might want to prevent the default behavior of an event.

```
1  import React from 'react'
2
3  function MyForm() {
4      let handleSubmit=(event)=>{
5          |      event.preventDefault()
6          |      alert('Form was submitted!!')
7      }
8      return (
9          <div>
10             <form onSubmit={handleSubmit}>
11                 <label>Name</label>
12                 <input type='text' name='name' />
13                 <button type='submit'>
14                     |      Submit
15                 </button>
16             </form>
17         </div>
18     )
19 }
20 export default MyForm
```

Using State with Events

```
1  import React, { useState } from 'react'
2
3  function MyInput() {
4      const [text, setText] = useState('')
5      let handleChange = (event) => {
6          |       setText(event.target.value)
7          |
8      }
9      return (
10         <div>
11             <input type='text' value={text} onChange={handleChange}/>
12             <p>You typed :{text}</p>
13         </div>
14     )
15 }
16 export default MyInput
```

Summary

- **Event names are camelCase:**

use `onClick` instead of `onclick`.

- **Event handlers are functions:**

pass the function reference directly not as a string.

- **Prevent default behavior:**

use `event.preventDefault()` to stop the default action of the event

- **Using state:**

update the state using the `useState` hook (in functional components)

Example

Welcome to employee registration page

Name

Phone

Email

Password

Confirm Password

Password doesn't match

Please Fill All the FIelds

Form Submitted Successfully

you, 2 months ago | 1 author (you)

```
1 import React from 'react'
2 import { useState } from 'react'
3 function StateExample7() {
4     const [formdata, setformdata] = useState({
5         username: '',
6         phone: '',
7         email: '',
8         password: '',
9         confirmpassword: ''
10    })
11    const [passwordError, setPasswordError] = useState(false)
12    const [FormFields, setFormFields] = useState(false)
13    const [Submit, setSubmit] = useState(false)
14    const handleChange = (event) => {
15        const name = event.target.name
16        const value = event.target.value
17        setformdata({ ...formdata, [name]: value })
18    }
19    const submitform = (e) => {
20        e.preventDefault()
```

```
20 e.preventDefault()
21 if(formdata.password !== formData.confirmpassword){
22     setPasswordError(true)
23     //setTimeout(window.location.reload(),9000)
24 }
25 else if(formData.name === '' || formData.phone === '' ||
26     formData.email === '' || formData.password === '' ||
27     formData.confirmpassword === '')
28 {
29     setFormFields(true)
30     //setTimeout(window.location.reload(),9000)
31 }
32 else{
33     setSubmit(true)
34     console.log("Name is ",formData.username)
35     console.log("Phone Number is ",formData.phone)
36     console.log("Email is ",formData.email)
37     console.log("Password is ",formData.password)
38     console.log("Confirm Password is ",formData.confirmpassword)
```

```
38 console.log("Confirm Password is ",formdata.confirmpassword)
39 }
40     }
41     return (
42     <div style={{display:"grid",placeItems:"center"}}>
43         <h2>Welcome to employee registration page</h2>
44         <div className='form-control'>
45             <form onSubmit={submitform}>
46                 <div>
47                     <label>Name</label><br></br>
48                     <input type='text' className='input'
49                         name="username"
50                         value={formdata.username}
51                         onChange={handleChange}/>
52                 </div>
53                 <div>
54                     <label>Phone</label><br></br>
```

```
54 |         <label>Phone</label><br></br>
55 | <input type='tel' className='input' name="phone"
56 | value={formData.phone} onChange={handleChange}/>
57 |     </div>
58 |     <div>
59 |         <label>Email</label><br></br>
60 | <input type='email' className='input' name="email"
61 | value={formData.email} onChange={handleChange}/>
62 |     </div>
63 |     <div>
64 |         <label>Password</label><br></br>
65 | <input type='password' className='input' name="password"
66 | value={formData.password} onChange={handleChange}/>
67 |     </div>
68 |     <div>
69 |         <label>Confirm Password</label><br></br>
70 | <input type='password' className='input' name="confirmpassword"
71 | value={formData.confirmpassword} onChange={handleChange}/>
72 |     </div>
```

```

71 value={formData.confirmpassword} onChange={handleChange}/>
72     </div>
73     <br></br>
74 {passwordError&&<p style={{color:"red",fontSize:"larger"}}>
75     *Password doesn't match*</p>}
76 {FormFields&& <p style={{color:"red",fontSize:"larger"}}>
77     *Please Fill All the FIELDS*</p>}
78 {Submit?<p style={{color:"green",fontSize:"larger"}}>
79     *Form Submitted Successfully*</p>:<p></p>}
80     <div>
81         <input type="submit"/>
82     </div>
83 </form>
84 </div>
85 </div>
86 )
87 }
88
89 export default StateExample7

```

Course Registration Form

Personal Information

Firstname:

Lastname:

Mobilenumber:

Email:

Password:

ConfirmPassword:

Confirm Password

DOB:

dd-mm-yyyy



Gender: ☐ Male ☐ Female ☐ Others

SelectState:



Course Information

SelectCourse:



DeliveryMode: ☐ Online ☐ Offline

Project

From Handling APP

- Create login form
- Capture input values
- Show submitted data on screen

Assignment

- Create signup form
- Show data after submit
- Create a password page form
- Show data after submit

Interview Questions

- What are synthetic events?
- Difference b/w html events and react events
- How to handle events in react